Roll no.....

Department of Physics
National Institute of Technology Kurukshetra

Programe: **B.Tech (All Branches)** | Exam: **FINAL (Regular)** | Date: **04-12-2025**
Subject: **Engineering Physics** | Course Code: **PHIC 101** | Duration: **3 hours**

Note:

- Attempt all the **Five questions**. Marks allotted for each question is shown on the right.
- Attempt **each** part of a particular question at the **same place** in your answer sheet.
- **Be precise** while writing the solutions. Wherever necessary, draw **relevant, neat** and **labelled schematic diagrams**.
- Any kind of communication device like phone, smart watch, pager, Bluetooth devices etc. are strictly prohibited.
- Scientifc calculator is allowed.

Relevant Physical constants: $h = 6.67 \times 10^{-34}$ J.s; $c = 3.0 \times 10^{8}$ m/s; $e = 1.6 \times 10^{-19}$ C; $m_e = 9.1 \times 10^{-31}$ kg; 1 eV$=1.6 \times 10^{-19}$ Joule; $k = 8.62 \times 10^{-5}$ eV/K; Avogadro no.$= 6.022 \times 10^{23}$

1.	**(a)** Discuss the conditions for a mathematical function to become a valid wavefunction? **[2]**
	(b) Write the differential forms of Maxwell's equations and discuss the relevance of Maxwell's corrections in Ampere's law? **[2 + 1]**
	(c) Define resistivity of a metal. Find the relaxation time of free electrons in a metal of resistivity 1.50×10^{-8} Ωm and having free electron density 5.5×10^{28} electrons m^{-3}. **[1 + 2]**
	(d) Write down the relatvisic kinectic energy expression and show that it reduces to classical kinectic energy expression for non-relativistic case? **[2]**
2.	**(a).** Derive the expressions of energy levels and wavefunctions for a particle confined in a 1 D box of width L. Calculate the energy diffence between the ground state and first excited states for a 3 D cubical box having a dimension 2 nm. **[4 + 3]**
	(b). A particle limited to the x axis has the wave function $\Psi = \mathbf{ax}$ between x = 0 and x = 1; and $\Psi = \mathbf{0}$ elsewhere. Find the probability in terms of **a** that the particle can be found between x = 0.45 and x =0.55. Also, estimate the expectation value of position for the particle in terms of **a**. **[3]**
	OR
	(a). Derive the complete expressions of Electromagnetic waves from the Maxwell's equations. **[6]**
	(b). The electric field component of an EM wave is given by, $\mathbf{E} = 80 \cos[10^8 t + \beta x]\hat{y}$ then calculate β. Calculate the time this EM wave takes to travel a distance λ/2. Calculate the wavelength of EM wave and determine the direction of the EM wave propagation. **[4]**
3.	**(a).** Define Atomic Packing Fraction for a typical cubic unit cell? Draw BCC and FCC unit cells and calculate the packing fraction for them. You can assume atoms are like hard spheres filling the unit cell. Which structure will show more resistivity for the electron flow? **[1 + 4 + 1]**
	(b). Copper exchibits FCC unit cell with lattice constant a = 3.61 Å. Calculate interplaner seperation for (2 0 0) and (2 2 0) Miller planes. Also, determine the angle between these two planes **[2 + 2]**

OR

(a). Explain the physical significance of density of states. Derive the expression for density of states for the electrons in a 2D metallic sheet. **[2 + 4]**

(b). Calculate the free electron concentration and Fermi energy in aluminum at 0K with the assumption that each aluminum atom contributes three electrons to the electron gas. The density and atomic mass of aluminum are 2.7 g/cm^3 and 27 amu respectively. **[4]**

4. **(a).** Derive the relation between Einstein's coeffiecients for a two level LASER system. Show that for this system, LASER action for wavelength 5000 Å is not possible at room temperature (300 K). **[4 + 2]**

(b). For a step index fiber, the core and cladding refractive indices are 1.54 and 1.5 respectively. Calculate following if the fiber is placed in the water (i) Critical angle, (ii) Fractional refractive index, (iii) Numerical aperture, and (iv) Acceptance angle in the water. **[4]**

5. **(a).** Derive the Lorentz transformation equation between two frames S and S' where frame S' is moving with the speed v along the positive x axis with respect to S frame. Show that it reduces to Galilean transformation equations in classical limits. **[6]**

(b). A spaceship that moves away from earth with the velocity $v/c = 0.85$ emits two light signals at the time interval $\Delta t' = 6s$. Here c is velocity of light.

i. What is the time interval ΔT (earth time) between the two signals arriving on earth? **[1]**

ii. What distance, measured from earth, did the spaceship covers in the duration of emitting the two light signals? **[1]**

iii. A body at rest in the spaceship has the mass $m_0 = 1$ kg. What is its kinetic energy measured from earth? **[2]**

OR

(a). Differentiate between nuclear fission and nuclear fusion process. Describe the construction and working principle of a Nuclear fission reactor with the help of neat schematic diagram. **[2 + 4]**

(b). Write short notes on (i) neutron cross-section and (ii) moderators. **[2 + 2]**

END